

MONTGOMERY COUNTY ROADWAY INSIGHTS

2019-2020 Traffic Collision Dashboard

PRESENTED BY



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DASHBOARD & KEY FINDINGS

Traffic Collision Analysis | 2019-2020

Objective: Identify crash trends and roadway safety patterns.



34,406

Total Crashes

Two-Year reporting period



2 Years

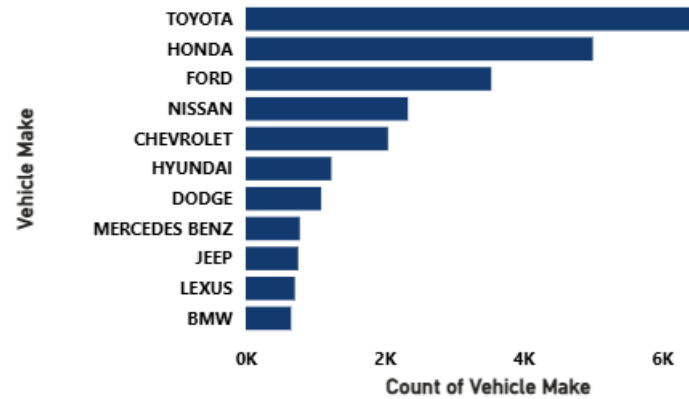
Reporting Period
2019-2020



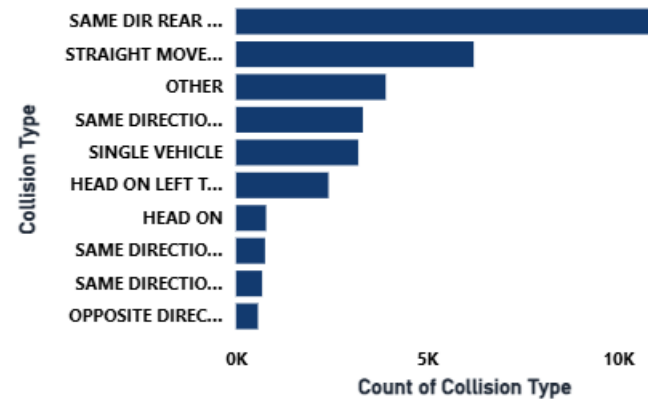
82%

Minor/ No Injury Cases
Share of total crashes

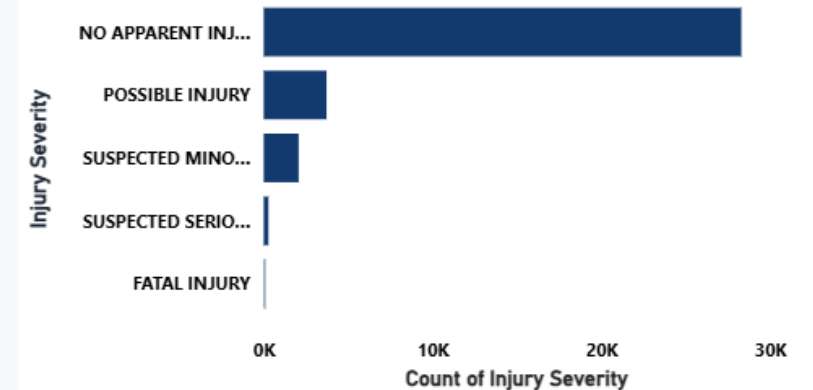
Top Vehicle Makes Involved in Crashes



Most Common Collision Types



Crash Injury Severity Distribution



KEY INSIGHTS



Toyota, Honda, and Ford accounted for the **highest share** of reported collisions, indicating concentration among **high-volume vehicle brands**.



Rear-end collisions were the most common collision type, representing the **largest proportion** of all reported crashes.



Crash severity remained **relatively low** overall, with the majority resulting in no apparent or minor injuries



Findings suggest opportunities to focus on **rear-end collision prevention** and continued driver awareness efforts.

RECOMMENDATIONS AND CONCLUSIONS

RECOMMENDATIONS



Increase focus on same-direction and rear-end collision prevention initiatives due to their high frequency within the dataset.



Use vehicle make and model trends to better anticipate claim exposure, reserve planning, and potential total loss frequency.



Continue monitoring injury severity trends to improve operational forecasting and claim workload management.



Use trend analysis to support faster decision-making and identify areas requiring additional risk evaluation.

CONCLUSION



This project demonstrated how crash reporting data can be transformed into meaningful operational insights through data cleaning, visualization, and trend analysis using Power BI.

| TOOLS & DATA PREPARATION



TOOLS USED:



Microsoft Excel



Power Query



Microsoft Power BI



DATA PREPARATION



Reviewed raw dataset structure



Removed duplicates and inconsistencies



Standardized column formatting



Cleaned and organized collision data



Prepared dashboard data for visualization and trend analysis

THANK YOU

QUESTIONS & DISCUSSION

PRESENTED BY



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